

Groundwater and population growth

Analysis map 05 documentation

Laudato Si Action Platform • Global Human Settlements • 4 October 2026

Science summary

The focus combines a positive groundwater-transgression indicator with positive population growth in the global upper quarter, 2015–2025. Recharge classification and aquifer salinity are retained separately as source categories.

Suggested local use: Investigate water demand, groundwater conditions, and opportunities to conserve water in local institutions.

Inputs and calculations

Derived field and unit	Source or calculation
GS_Groundwater Groundwater transgression, 2025 (source event indicator)	Water / NS_ESB_GWA_2025
GS_PopGrowth Population change, 2015–2025 (%)	$100 \times (\text{Pop25} - \text{Pop15}) / \text{Pop15}$

Focus when Groundwater > 0 and PopGrowth > 0 and $\geq 13.65561811911165\%$. Population growth uses the global upper quartile. GS_Recharge and GS_Salinity retain WA_GWR_MMA_2025 and WA_GWS_SAL_2025 categories separately.

Screening and missing observations

Null, nonnumeric, nonfinite and sentinel values ≤ -999 are excluded. Negative values are excluded for nonnegative measures, but signed indices and changes can remain negative. Percentages must be within 0–100; heat days within 0–366. Percentage changes require a positive baseline. Compound screening uses three-valued logic: AND is false if any component is false, true if all are true, otherwise unknown; OR is true if any is true, false if all are false, otherwise unknown.

Quartiles use sorted nearest rank at $\text{ceil}(N \times q) - 1$, including ties; theme 09 instead interpolates at $(N - 1) \times q$. Thresholds calibrated on connected cities are applied to all cities for context. Focus display layers in maps 02, 03, 04 and 07 also require $\text{GS_MemberN} > 0$.

Results and comparison groups

1,209 focused cities among 11,422 classifiable global cities; 60 focused cities among the 638 connected cities; 223 matched member point records inherit a focused-city classification. These are different units, not equivalent enrolment counts.

Layer	Valid primary value	Unavailable primary	Records with positive focus
Cities	11422	0	1209
Members	1929	810	223
Dioceses	2252	622	403
Conferences	120	2	71
Members20km	2372	367	299
Members100km	2666	73	617

The primary field is GS_Groundwater. Aggregate “positive focus” means at least one focused city among classifiable cities; it does not indicate that the territory as a whole passes the city screen.

Spatial scope

Cities use whole mapped urban centres. Members inherit values only after one unique city match: 1,929 of 2,739 point records match 638 cities; 810 unmatched records keep null indicators and zero city-linked counts. Dioceses, conferences and member buffers use an equal-city mean of distinct intersecting cities, separately for each metric’s valid observations. These are neither population-weighted nor area-weighted territorial estimates. A cross-boundary city can occur in multiple summaries, so totals across overlapping territories are not additive.

Rankings and percentiles

Rank 1 is the highest numeric value, with competition ranks for ties. Percentile is $100 \times (\text{number below} + \text{half the number equal}) / \text{valid N}$. Higher percentile means a higher numeric value, not better or worse conditions. Missing values are excluded. All-equal values have rank 1 and percentile 50.

Cities are compared globally once each. Member points inherit their city’s global rank rather than creating repeated observations. Dioceses, conferences, 20 km buffers and 100 km buffers each have a separate comparison group of valid equal-city means. Rankings of totals or percentages must retain the field’s unit and reference year.

Fields for popups and the sidebar

Field	Meaning
GS_Short05	Concise prepared popup summary
GS_Long05	Longer sidebar narrative with methods, caveats and ecclesiastical context
GS_Ecclesial	Readable ecclesiastical context
GS_Focus05	City/member 0, 1 or null; aggregate focused-city percentage or null
GS_CityN / GS_MemberN / GS_RegN	City count / linked point records / distinct nonblank registration IDs
GS_DirectN	Member points directly in a jurisdiction; different from city-linked counts

For each metric GS_X, GS_X_Rank is descending competition rank, GS_X_Pctile the numeric-value percentile, GS_X_RankN the valid comparison group and GS_X_CityN the contributing valid city count. Keep descriptive aliases visible. Connect a sidebar to the selected feature's GS_Long05; the project provides text, not a published web application.

Ecclesiastical information

All 43 diocesan and 186 conference original attributes were preserved and compared value-for-value. Names, rites, traditions, hierarchy, source dates, population, Catholic statistics, clergy, parishes, leadership, enrolment, education and duplicate source series remain available. Member and buffer source attributes and geometry also remain. GS_Ecclesial provides readable context in longer summaries. Leadership and historic statistics retain source dates; they were not revalidated as current officeholders. Fields with suffix _26 and other original series were not merged or silently overwritten.

Quality control and evidence

Prior GroundwaterGrowth comparison verified population growth and focus for 11,422 cities.

Saved calculations and valid city counts were checked after writing. All 28 metric rankings were independently verified for cities, dioceses, conferences, 20 km buffers and 100 km buffers. Aggregate means were reproduced from saved relationship keys. Original geometry and attributes were compared for the five member and ecclesiastical foundations. This validates the implemented calculations, not the accuracy of every publisher indicator or a new independent spatial-overlay algorithm.

Project checks found 12 analysis maps among 15 total maps, no broken layers and 88 saved popup checks passed. These checks do not certify an ArcGIS Online or Experience Builder deployment. Evidence:

../13_Source_Data_Documentation/Evidence/analysis_qc.json, independent_validation.json, relationship_qc.json and final_project_validation.json.

Example from the saved project

Apia

Groundwater transgression, 2025 (source event indicator): 0.0 source event indicator; Population change, 2015–2025 (%): 14.8 %. Does not meet the relative screening condition. For Groundwater transgression, 2025 (source event indicator), descending rank 5,816 of 11,422 global urban centres; numeric-value percentile 24.5.

Groundwater transgression, 2025 (source event indicator): 0.0 source event indicator; Population change, 2015–2025 (%): 14.8 %. Does not meet the relative screening condition. For Groundwater transgression, 2025 (source event indicator), descending rank 5,816 of 11,422 global urban centres; numeric-value percentile 24.5. Indicators refer to this mapped urban centre. 0 member point records and 0 distinct nonblank registration IDs are linked through intersecting cities. Repeated registration IDs are distinguished from point records. The focus combines a positive groundwater-transgression indicator with positive population growth in the global upper quarter, 2015–2025. Recharge classification and aquifer salinity are retained separately as source categories. Rank 1 is the highest numeric value, with competition ranks for ties. Percentile is $100 \times (\text{number below} + \text{half the number equal}) / \text{valid N}$. Higher percentile means a higher numeric value, not better or worse conditions. Missing values are excluded. All-equal values have rank 1 and percentile 50. Possible local response: Investigate water demand, groundwater conditions, and opportunities to conserve water in local institutions.

Reproducibility and interpretation

Open `../LSAP_Global_Human_Settlements.aprx` and select analysis map 05. The same six main analysis datasets supply maps 01–11 with different themed fields and popup configurations. Full source copies and the earlier UCDB reference layer are retained in source and foundation maps. Local country outlines and the Equal Earth display projection support the map context.

Source reference years can include modelled estimates. City context does not diagnose an individual address, Catholic property or service outcome. The suggested responses are starting points for local discussion, not validated intervention priorities. All numeric rankings remain descriptive.

Reproduce using the archived workflow in `../13_Source_Data_Documentation/Workflow` and consult the complete `Field_Dictionary.csv`. GIS execution requires ArcGIS Pro with `arcpy` and valid source paths. The project retains external CorrectGeo reference dependencies and is not a standalone portable package.